

Week 1: Econ 645 Introduction

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Chapter 1

Overview

Welcome to Econ 645. We will cover how to implement Stata commands for econometrics, along with data management. Data management will be key for transparent and reproducible work or research. For tonight, we will cover the following topics before delving into Mitchell and Wooldridge:

- Julian Reif's Coding Guide Folder Structure
- Executing do-files and making log files
- Automate Graphs
- Helpful Advanced Techniques
- Publish your results

Chapter 2

Coding Guide for Folder Structure

Please see Julian Reif's coding guide at <https://julianreif.com/guide/>. These are very good tips very about replicability and organization. Having your files organized in a folder as an example below will not only with organization, but with replication, as well. You can hand over the entire folder structure, and another users will be able to run the master do file with just changing the main folder location once.

Folder Structure

- analysis/
 - data/
 - processed/
 - results/
 - * figures/
 - * tables/
 - * scripts/
 - 1_process_raw_data.do
 - 2_...
 - run.do
- paper/
 - manuscript.tex
 - figures/
 - tables/

Chapter 3

Executing do-files and making log files

From Mitchell 10.3, you should always use a do file when you work with Stata. I find log files helpful, but do files are essential. Do files provide transparency to show your work, and it help provide replication. One key feature of do files are comments. Comments are an essential part for two reasons. It will help yourself when you go back to your own code to describe what is going one. Second, it helps with replication, since someone should be able to run your code top to bottom and get the same results that you have in your paper, report, or document.

Never use point and click. It is available in Stata, but don't lower yourself to a SPSS standard.

In case you are new to do files. We'll start off with a simple example. We'll pull some data, summarize some data, and tabulate some data.

```
use wws.dta, clear
summarize age wage hours
tabulate married
```

/Users/Sam/Desktop/Econ 645/Data/Mitchell

(Working Women Survey)

Variable	Obs	Mean	Std. Dev.	Min	Max
age	2,246	36.25111	5.437983	21	83
wage	2,246	288.2885	9595.692	0	380000
hours	2,242	37.21811	10.50914	1	80

married	Freq.	Percent	Cum.
0	804	35.80	35.80
1	1,442	64.20	100.00
Total	2,246	100.00	

We can look at a similar file called example1.do

```
type example1.do
```

We can actually run a do file within a do file

```
do example1.do
```

(Working Women Survey w/fixes)

Variable	Obs	Mean	Std. Dev.	Min	Max
age	2,246	36.22707	5.337859	21	48
wage	2,244	7.796781	5.82459	0	40.74659
hours	2,242	37.21811	10.50914	1	80
married	Freq.	Percent	Cum.		
0	804	35.80	35.80		
1	1,442	64.20	100.00		
Total	2,246	100.00			

You can use the doedit command if you want to open the Do-file Editor, but you can just click to open the Do-File (this is an exception from what I said above).

Since we already have the Do-File Editor open, running the doedit will just open a new blank do file.

```
doedit
```

Note: you can write do files in Notepad or TextEdit, but you don't get any of the benefits, such as highlighted text.

Let's look at a do file that uses the log command

```
type example2.do
```

Let's run it.

```
log using example2
use wws2, clear
summarize age wage hours
tabulate married
log close example2
```

Note: It is important to note that when opening a log it must end with a “log close” command. If your do files bombs out (fails to finish) before reaching the log close command, your log will remain open.

Even if we close the log file, we'll get an error if we try to run the do file again. So we need to add the replace option.

```
log using example2, replace
use wws2, clear
summarize age wage hours
tabulate married
log close
```

```
name: <unnamed>
log: /Users/Sam/Desktop/Econ 645/Data/Mitchell/example2.smcl
log type: smcl
opened on: 10 Aug 2025, 09:41:12
```

(Working Women Survey w/fixes)

Variable	Obs	Mean	Std. Dev.	Min	Max
age	2,246	36.22707	5.337859	21	48
wage	2,244	7.796781	5.82459	0	40.74659
hours	2,242	37.21811	10.50914	1	80
married	Freq.	Percent	Cum.		
0	804	35.80	35.80		
1	1,442	64.20	100.00		
Total	2,246	100.00			

```
name: <unnamed>
```

```
log: /Users/Sam/Desktop/Econ 645/Data/Mitchell/example2.smcl
log type: smcl
closed on: 10 Aug 2025, 09:41:12
```

Chapter 4

Automate Graphs and Figures

Stata makes saving graphs easy, so if there is a minor fix all you need to do is rerun the scripts and have your graph fixed. Use the **graph export** command to save a graph.

```
*Get CPS Data
use jun25pub.dta
gen earnings=pternwa/100 if prrelg==1
histogram earnings if prrelg==1, title("Wages in June 2025") caption("Source: Current Population
graph export "Wages_June_2025.png"
```

```
/Users/Sam/Desktop/Econ 645/Data/CPS
```

```
(28,617 missing values generated)
```

Variable	Obs	Mean	Std. Dev.	Min	Max
earnings	9,939	1596.086	2184.658	0	13145.81

```
(bin=39, start=0, width=337.07205)
```

```
option replace not allowed
```

```
r(198);
```

```
r(198);
```

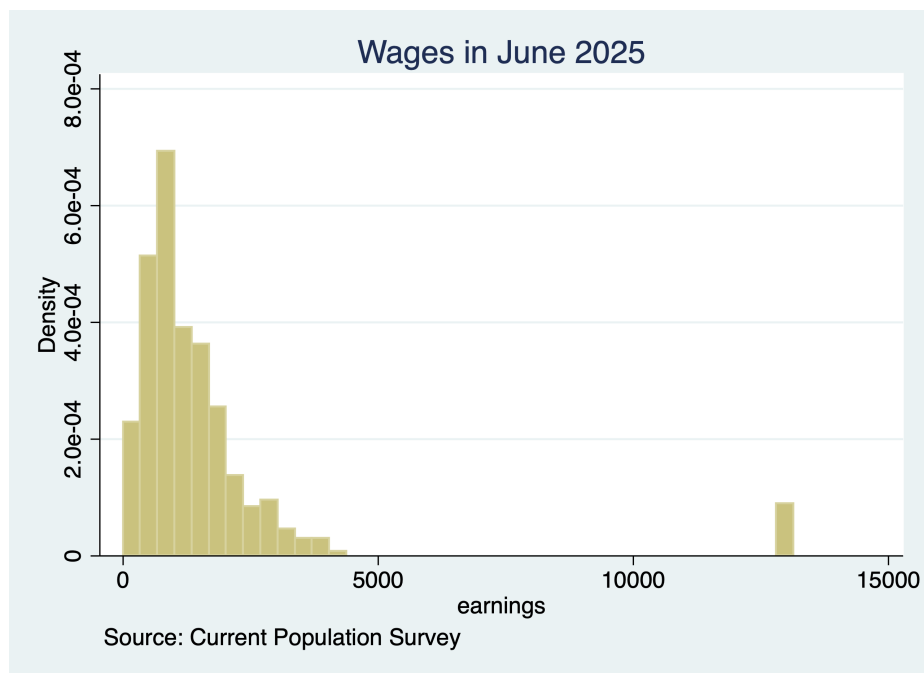


Figure 4.1: Wages Histogram

Chapter 5

Stata Programming Techniques Helpful for Econ 645

A brief overview of techniques that will be helpful for completing assignments and empirical project topics include - Local Macros, Loops, Tempfiles, By Sort and Egen, and Weights<

1. Macros
2. Local Macros
3. Global Macros
4. Loops
5. Dataframes and Tempfiles
6. Appending
7. Bysort and EGEN
8. Survey Weights
9. Accessing results in Stata's stored memory

5.1 Stata Programming: Macros

One of the essential parts of Stata programming are macros. Macros variables have many uses and can contain strings or numerics. We can use them in loops, in regressions, in calling temporary files, etc. There are two kinds of macros: local and global.

5.2 Local Macros

Local macros work within a single executed do file and removed once the session is run. If you are running local macros you must run the line of code that initializes the macro and the command that utilizes it.

You need to initialize the local macro E.g.: `local i = 1` *Key:* You need to call the macro with `key` (or left quote key just above tab key) and `'` key. E.g.: `display "i'"` *E.g.: `replace x = 0 if y = 'i'`

```
local i = 1
display "`i'"
```

1

```
local k = `i' + 1
display "`k'"
```

2

5.3 Global Macros

Global macros are variables that stays within memory even after a new session is run. These macro variables can work across multiple do files, as well. You need to initialize a global macro E.g.:

```
global j = 2
```

Next, you call the macro with `$`

```
display "$j"
```

2

You can use local and global macro variables for qualifiers, as well

```
replace z = 1 if y = $i
```

Another great use of macros is to find a list of for local levels of a categorical variable, and store it in macro variable


```
levelsof varname, local(levels)
foreach l of local levels {
    command if varname == `l'
}
```

We'll demo this later, but it is quite useful

You can also use local macros to test regression models We'll demo this later, as well

5.4 Looping

*Looping has many uses when you want to apply a function or command over a repeated set of values There is forvalues and foreach - I typically use foreach given it's versatility but Stata says that forvalues can be more efficient *E.g.: Loop over years and iterate i*

5.4.1 Loop over a number

```
local i = 0
foreach num of numlist 2015/2019 {
    display "`num'"
    local i = `i'+1
    display "This is the `i' loop"
}
```

```
2015
This is the 1 loop
2016
This is the 2 loop
2017
This is the 3 loop
2018
This is the 4 loop
2019
This is the 5 loop
```

5.4.2 Loop over a local macro

*E.g.: Loop over months and years to read in new files

```

local month jan feb mar apr may jun jul aug sep oct nov dec
foreach y of numlist 2018/2019 {
    foreach m of local month {
        local filename = "`m'`y'.dta"
        display "`filename'"
    }
}

```

```

jan2018.dta
feb2018.dta
mar2018.dta
apr2018.dta
may2018.dta
jun2018.dta
jul2018.dta
aug2018.dta
sep2018.dta
oct2018.dta
nov2018.dta
dec2018.dta
jan2019.dta
feb2019.dta
mar2019.dta
apr2019.dta
may2019.dta
jun2019.dta
jul2019.dta
aug2019.dta
sep2019.dta
oct2019.dta
nov2019.dta
dec2019.dta

```

5.5 Data Frames and Tempfiles

Data frames are a relatively new part of Stata (Stata 16+) that let you switch between datasets. Before you could only work on one dataset at a time. You can read an introduction to Stata data frames from Asjad Naqvi.

For older versions of Stata, we have the tempfiles workaround. Tempfiles are useful since they are a bypass around an older Stata limitation of one dataframe at a time. Later iterations of Stata introduced the dataframe functionality, but the tempfile method still works well. We will append three years of CPS MORG

data from NBER and generate short 1-year panels. *Note:* Tempfiles are really just macros.

5.6 Appending multiple CPS files

Get CPS Data

Small CPS Files have only a few variables in them compared to the usual. The full Census CPS micro data files found at: <https://www.census.gov/data/datasets/time-series/demo/cps/cps-basic.html>.

Set up initial tempfile so we can add each year of CPS data. Save the macro and set it to emptyok.

```
tempfile cps
save `cps', emptyok
```

Create a local variable to loop over each year and month to append monthly CPS files into 1 cps file.

```
cd "/Users/Sam/Desktop/Data/CPS"
*Set month macro
local month jan feb mar apr may jun jul aug sep oct nov dec
local filecount = 0

tempfile cps
save `cps', emptyok

foreach y of numlist 23/24 {

    foreach m of local month {

        *Show the year and month
        display "`m'`y'"
        local filename "small`m'`y'pub.dta"
        display "`filename'"

        *Open the monthly data file
        use "`filename'", clear

        *Append monthly data file to cps tempfile
        quietly append using `cps'

        save `cps', replace
        clear
    }
}
```

```

        *Count the number of monthly files appended
        local filecount = `filecount' + 1
    }
}
*Get the appended CPS File
use `cps'
tab hrmonth hryear4
save "cps23_24.dta", replace

```

/Users/Sam/Desktop/Data/CPS

```

(note: dataset contains 0 observations)
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_36840.000001 saved

jan23
smalljan23pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_36840.000001 saved
feb23
smallfeb23pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_36840.000001 saved
mar23
smallmar23pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_36840.000001 saved
apr23
smallapr23pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_36840.000001 saved
may23
smallmay23pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_36840.000001 saved
jun23
smalljun23pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_36840.000001 saved
jul23
smalljul23pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_36840.000001 saved
aug23
smallaug23pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_36840.000001 saved
sep23
smallsep23pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_36840.000001 saved

```

```
oct23
smalloct23pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_36840.000001 saved
nov23
smallnov23pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_36840.000001 saved
dec23
smalldec23pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_36840.000001 saved
jan24
smalljan24pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_36840.000001 saved
feb24
smallfeb24pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_36840.000001 saved
mar24
smallmar24pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_36840.000001 saved
apr24
smallapr24pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_36840.000001 saved
may24
smallmay24pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_36840.000001 saved
jun24
smalljun24pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_36840.000001 saved
jul24
smalljul24pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_36840.000001 saved
aug24
smallaug24pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_36840.000001 saved
sep24
smallsep24pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_36840.000001 saved
oct24
smalloct24pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_36840.000001 saved
nov24
smallnov24pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_36840.000001 saved
dec24
smalldec24pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_36840.000001 saved
```

HRMONTH	HRYEAR4		Total
	2023	2024	
1	125,982	126,802	252,784
2	124,754	126,784	251,538
3	124,477	124,581	249,058
4	126,798	126,850	253,648
5	127,559	126,945	254,504
6	127,491	126,112	253,603
7	127,424	126,158	253,582
8	127,930	127,031	254,961
9	126,665	126,590	253,255
10	127,941	126,387	254,328
11	126,917	126,686	253,603
12	126,832	126,743	253,575
Total	1,520,770	1,517,669	3,038,439

file cps23_24.dta saved

CPS Basic Data Dictionary can be found at: [CPS Basic Data Dictionary](#).

Check all years are there

5.7 By Sort and EGEN

Please note that we will get into `bysort egen` later in the semester. However, `bysort: egen` is a powerful combination that it is worth bring up now. You can summarize, replace, or create new variables by multiple groups with the `by sort` and `egen` commands. Let's generate `laborforce`

1. 1 is employed at work;
2. 2 is employed absent;
3. 3 is unemployed layoff;
4. 4 is unemployed looking;
5. 5 is NILF retired;
6. 6 is NILF disabled; and
7. 7 is NILF other

```
gen laborforce = .
replace laborforce = 0 if pemplr >= 5 & pemplr <= 7
replace laborforce = 1 if pemplr >= 1 & pemplr <= 4
label define laborforce1 0 "NILF" 1 "Labor Force"
```

```

label values laborforce laborforce1
tab laborforce

gen employed = .
replace employed = 0 if pemplr >= 3 & pemplr <= 7
replace employed = 1 if pemplr >= 1 & pemplr <= 2
label define employed1 0 "Not Employed" 1 "Employed"
label values employed employed1
tab employed

```

Generate a race/ethnicity category from existing

```

gen race_ethnicity = .
replace race_ethnicity = 1 if ptdtrace == 1 & pehspon == 2
replace race_ethnicity = 2 if ptdtrace == 2 & pehspon == 2
replace race_ethnicity = 3 if pehspon == 1
replace race_ethnicity = 4 if ptdtrace == 3 & pehspon == 2
replace race_ethnicity = 5 if (ptdtrace == 4 | ptdtrace == 5) & pehspon == 2
replace race_ethnicity = 6 if (ptdtrace >= 6 & ptdtrace <= 26) & pehspon == 2
label define race_ethnicity1 1 "White NH" 2 "Black NH" 3 "Hispanic or Latino/a" ///
4 "Native American NH" 5 "Asian or Pacific Islander NH" 6 "Multiracial NH"
label values race_ethnicity race_ethnicity1
tab race_ethnicity

```

Sort by Sex

```

sort pesex
*Summarize laborforce by sex
by pesex: sum laborforce
bysort: sum laborforce

```

Sort by Sex and Race.

```

sort pesex race_ethnicity
*Summarize laborforce by sex and race
by pesex race_ethnicity: sum laborforce
bysort pesex race_ethnicity: sum laborforce

```

Generate Age Bin to find individuals over 16.

```

gen over_16 = .
replace over_16 = 0 if prtage < 16
replace over_16 = 1 if prtage >= 16

```

```
label define over_16a 0 "Under 16" 1 "16 and older"
label values over_16 over_16a
tab over_16
```

Generate unweighted laborforce participation rate for each group with `by sort` and `egen`. `Bysort` works, as well, but I usually use `sort` on one line and `by` on the other

```
sort pesex race_ethnicity
by pesex race_ethnicity: egen mean_lfpr = mean(laborforce) if over_16 == 1
by pesex race_ethnicity: sum mean_lfpr
```

```
*Counting and indexing/subscripting within groups
gen idcount = .
by pesex race_ethnicity: replace idcount = _n
gen idcount2 = .
by pesex race_ethnicity: replace idcount2 = idcount[_N]
```

5.8 Retrieving estimates stored in Stata's memory

Instead manually entering summary or regression results, we can use scalars in Stata's memory to retrieve the information. This reduces human error of copying and pasting.

For summary statistics we can use the **return list** command.

```
cd "/Users/Sam/Desktop/Data/CPS"
use smalljan24pub.dta, clear
sum pternwa if prrelg==1

return list

gen demean_earnings=pternwa-`r(mean)' if prrelg==1
summarize demean_earnings
```

```
/Users/Sam/Desktop/Data/CPS
```

Variable	Obs	Mean	Std. Dev.	Min	Max
-----+-----					
pternwa	10,666	137274.8	146839.5	0	1125627

scalars:

```

      r(N) = 10666
r(sum_w) = 10666
r(mean) = 137274.7503281455
r(Var) = 21561833661.67427
r(sd) = 146839.4826389492
r(min) = 0
r(max) = 1125627
r(sum) = 1464172487

```

(116,136 missing values generated)

Variable	Obs	Mean	Std. Dev.	Min	Max
demean_ear~s	10,666	-3.79e-12	146839.5	-137274.8	988352.2

We can also grab information about a regression

```

cd "/Users/Sam/Desktop/Data/CPS"
use smalljan24pub.dta, clear
gen llearn=ln(pternwa) if prerelg==1
gen exp=prtage-16
replace exp=0 if exp==-1

reg llearn c.exp##c.exp i.peeduca i.peernlab if prerelg==1

ereturn list

matrix list e(b)

*Coefficients and standard errors
display _b[1.peernlab] _se[1.peernlab]

```

/Users/Sam/Desktop/Data/CPS

(116,150 missing values generated)

(27,667 missing values generated)

(1,209 real changes made)

Source	SS	df	MS	Number of obs	=	10,652
-----+-----				F(18, 10633)	=	238.75

Model		2310.98949	18	128.388305	Prob > F	=	0.0000
Residual		5717.80316	10,633	.537741292	R-squared	=	0.2878

Total		8028.79265	10,651	.753806465	Adj R-squared	=	0.2866
					Root MSE	=	.73331

lnearn		Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	

exp		.0654986	.0019087	34.31	0.000	.0617571	.0692401
c.exp#c.exp		-.0010166	.0000324	-31.38	0.000	-.0010801	-.0009531
peeduca							
32		.2151811	.2348611	0.92	0.360	-.2451906	.6755528
33		.1297205	.2243499	0.58	0.563	-.3100474	.5694884
34		.1947927	.2240222	0.87	0.385	-.2443328	.6339181
35		-.0479483	.2153955	-0.22	0.824	-.4701637	.3742672
36		-.1032521	.2123809	-0.49	0.627	-.5195583	.3130541
37		.0602974	.2103972	0.29	0.774	-.3521206	.4727154
38		.1407128	.2128599	0.66	0.509	-.2765325	.5579581
39		.4391957	.203923	2.15	0.031	.0394684	.838923
40		.4534936	.2043117	2.22	0.026	.0530045	.8539827
41		.5707234	.2059659	2.77	0.006	.1669917	.974455
42		.5189681	.2053085	2.53	0.011	.1165251	.9214112
43		.914999	.2038914	4.49	0.000	.5153337	1.314664
44		1.025085	.2044413	5.01	0.000	.6243415	1.425828
45		1.463769	.2108507	6.94	0.000	1.050462	1.877076
46		1.3309	.2076626	6.41	0.000	.9238427	1.737958
2.peernlab		-.0496265	.024125	-2.06	0.040	-.0969159	-.0023371
_cons		10.07725	.2065816	48.78	0.000	9.672309	10.48219

scalars:

```

e(N) = 10652
e(df_m) = 18
e(df_r) = 10633
e(F) = 238.7547824846974
e(r2) = .2878377352595289
e(rmse) = .7333084563678371
e(mss) = 2310.989494457061
e(rss) = 5717.803159756108
e(r2_a) = .2866321563292809
e(ll) = -11800.89312136709
e(ll_0) = -13608.80112438194

```

```

e(rank) = 19

macros:
    e(cmdline) : "regress llearn c.exp##c.exp i.peeduca i.peernlab if prerelg==1"
    e(title) : "Linear regression"
    e(marginsok) : "XB default"
    e(vce) : "ols"
    e(depvar) : "llearn"
    e(cmd) : "regress"
    e(properties) : "b V"
    e(predict) : "regres_p"
    e(estat_cmd) : "regress_estat"

matrices:
    e(b) : 1 x 21
    e(V) : 21 x 21

functions:
    e(sample)

e(b) [1,21]
           c.exp#      31b.      32.      33.      34.      35.      3
           exp      c.exp      peeduca      peeduca      peeduca      peeduca      peeduca      peeduca
y1      .06549863  -.00101659      0      .21518108      .12972052      .19479266      -.04794826      -.103252
           38.      39.      40.      41.      42.      43.      44.      4
           peeduca      peeduca      peeduca      peeduca      peeduca      peeduca      peeduca      peeduca
y1      .1407128      .43919568      .4534936      .57072336      .51896812      .91499896      1.0250848      1.46376
           1b.      2.
           peernlab      peernlab      _cons
y1      0      -.04962649      10.077247

00

```

5.9 Survey Weights

We can use the Current Population Survey public use microdata and replicate published BLS numbers. We need to adjust our numbers with the appropriate weights in order to replicate the published numbers.

First, we need to adjust weights the weights. In the CSV files the weights need to be divided by 1000 since the documentation says implies 4 decimals.

Composite weights *cmpwgt2** are used for final BLS tabulations of labor force

```
gen cmpwgt2 = pwcmpwgt/10000
```

Outgoing Rotation Weights *pworwgt* are used for earnings only person in interviews 4 and 8

```
gen orwgt2 = pworwgt/10000
```

PWSSWGT weights are used for general tabulations of sex, race, states, etc.

```
gen sswgt2 = pwsswgt/10000
```

PWVETWGT are used to study the veteran population

```
gen vetwgt2 = pwvetwgt/10000
```

PWLWGT are weights used to study someone over multiple CPS interviews

```
gen lgwgt2 = pwlgwgt/10000
```

If we are appending all months, we need to divide by 12 for the Basic CPS to get annual weights.

```
replace cmpwgt2 = cmpwgt2/12
```

If we are appending all months across multiple years, we need to get a composite weight for all of the CPS files

```
cd "/Users/Sam/Desktop/Data/CPS"
use cps23_24.dta
gen cmpwgt3 = pwcmpwgt/24
```

Use the `SvySet` command to set up the survey design.

```
svyset [pw=cmpwgt3]
```

Use the `svy:` command vars to utilize the survey design.

```
svy: tab employed hyear4, count cellwidth(20) format(%20.2gc)
```

```
cd "/Users/Sam/Desktop/Data/CPS"
use cps23_24.dta
gen cmpwgt3 = pwcmpwgt/10000
replace cmpwgt3 = cmpwgt3/12
svyset [pw=cmpwgt3]
gen employed = .
replace employed = 0 if pmlr >= 3 & pmlr <= 7
replace employed = 1 if pmlr >= 1 & pmlr <= 2
label define employed1 0 "Not Employed" 1 "Employed"
label values employed employed1
tab employed
svy: tab employed hyear4, count cellwidth(20) format(%20.2gc)
```

```
/Users/Sam/Desktop/Data/CPS
```

```
(655,995 missing values generated)
```

```
(1,946,105 real changes made)
```

```
    pweight: cmpwgt3
      VCE: linearized
Single unit: missing
  Strata 1: <one>
      SU 1: <observations>
    FPC 1: <zero>
```

```
(3,038,439 missing values generated)
```

```
(838,080 real changes made)
```

```
(1,138,576 real changes made)
```

employed	Freq.	Percent	Cum.
-----+-----			
Not Employed	838,080	42.40	42.40
Employed	1,138,576	57.60	100.00
-----+-----			
Total	1,976,656	100.00	

```
(running tabulate on estimation sample)
```

```
Number of strata   =           1           Number of obs   =   1,976,656
```

Number of PSUs	= 1,976,656	Population size	= 535,513,623
		Design df	= 1,976,655

Chapter 6

Displaying your results with publication-quality tables with `estout` and `outreg`

Stata has two suites of functions that help produce publication-quality tables: `estout` and `outreg2`. While it is easy to copy and paste tables, this is not a best practice. You want to have your Stata scripts reduce the amounts of pointing-and-clicking. Pointing-and-clicking is highly repetitive and more prone to user error. It is better to avoid doing this! Using `estout` or `outreg2` will help you in the long-run.

6.1 Estout

`Estout` and its functions are a way to produce publication-quality tables of descriptive statistics or regression results. The main function will be *esttab*.

You can install `estout` by running the following `ssc` install command: <http://repec.sowi.unibe.ch/stata/estout/index.html>

```
ssc install estout
```

`Esttab` is the main function in `estout` that provides publication-quality tables. The basics of `esttab` are the following:

```
esttab [ namelist ] [ using filename ] [ , options estout_options ]
```

6.1.1 Esttab Examples

Here is an example of a simple regression output comparing two models. We use `eststo` to store the model and use `esttab` to display the results.

```
quietly sysuse auto
eststo m1: quietly regress price weight mpg
eststo m2: quietly regress price weight mpg foreign
esttab m1 m2
```

	(1)	(2)
	price	price
weight	1.747** (2.72)	3.465*** (5.49)
mpg	-49.51 (-0.57)	21.85 (0.29)
foreign		3673.1*** (5.37)
_cons	1946.1 (0.54)	-5853.7 (-1.73)
N	74	74

t statistics in parentheses
 * p<0.05, ** p<0.01, *** p<0.001

We can replace t-statistics with p-values with “p”. Or, we can use “se” for standard errors, and we can use ci “ci” for confidence interval. We can add scalars to the model output as well. We can use “F” to display the overall F-statistic, “df_r” for degrees of freedom, and “rmse” for root mean squared error. We can also use “r2” for R-Squared and “ar2” for adjusted R-Squared.

```
quietly sysuse auto
eststo m1: quietly regress price weight mpg
eststo m2: quietly regress price weight mpg foreign
esttab m1 m2, p scalars(F df_r rmse) r2 ar2
```

	(1)	(2)
	price	price

weight	1.747** (0.008)	3.465*** (0.000)
mpg	-49.51 (0.567)	21.85 (0.769)
foreign		3673.1*** (0.000)
_cons	1946.1 (0.590)	-5853.7 (0.087)

N	74	74
R-sq	0.293	0.500
adj. R-sq	0.273	0.478
F	14.74	23.29
df_r	71	70
rmse	2514.0	2130.8

p-values in parentheses

* p<0.05, ** p<0.01, *** p<0.001

With the `title()` option, we can add a title to the table output. We can also change the model names at the top of the columns with the `mtitle()` option. We can move the p-values, t-statistics, or standard errors to the right of the beta coefficients with the option “wide”. We can also add a note with the `addnote()` option

```
quietly sysuse auto
eststo m1: quietly regress price weight mpg
eststo m2: quietly regress price weight mpg foreign
esttab m1 m2, title("Regression Analysis Example") p scalars(F df_r rmse) r2 ar2 mtitle("Model 1")
```

Regression Analysis Example

	(1)		(2)	
	Model 1		Model 2	

weight	1.747**	(0.008)	3.465***	(0.000)
mpg	-49.51	(0.567)	21.85	(0.769)
foreign			3673.1***	(0.000)
_cons	1946.1	(0.590)	-5853.7	(0.087)

N	74		74	
R-sq	0.293		0.500	

adj. R-sq	0.273	0.478
F	14.74	23.29
df_r	71	70
rmse	2514.0	2130.8

p-values in parentheses
 * p<0.05, ** p<0.01, *** p<0.001

6.1.2 Exporting into Word, Excel, or LaTeX format

****Do not copy and paste! Automate!***

We can use `esttab` with the export options to export into an RTF (Word), Excel, or LaTeX format.

6.1.2.1 Export into Excel format

If you want to use your table in Excel. Don't copy and paste! Just add "using" after `esttab` and include the ".csv" file format to the file name

```
sysuse auto
eststo m1: quietly regress price weight mpg
eststo m2: quietly regress price weight mpg foreign
esttab m1 m2 using example.csv, title("Regression Analysis Example") p scalars(F df_r r
```

6.1.2.2 Export into Word format

If you want to add your table to a Word document, add "using" after `esttab` and include the ".rtf" file format to the file name.

```
sysuse auto
eststo m1: quietly regress price weight mpg
eststo m2: quietly regress price weight mpg foreign
esttab m1 m2 using example.rtf, title("Regression Analysis Example") p scalars(F df_r r
```

6.1.2.3 Export into LaTeX format

You can also export your tables into a LaTeX table. You can export your file and load it directly into Overleaf. Add "using" after `esttab` and include the ".tex" file format to the file name

```
sysuse auto
eststo m1: quietly regress price weight mpg
eststo m2: quietly regress price weight mpg foreign
esttab m1 m2 using example.tex, title("Regression Analysis Example") p scalars(F df_r rmse) r2 ar
```

6.2 Outreg2

Outreg2 is another set of functions that help produce publication-quality tables. My preference is for esttab in estout, so I will provide you with a reference to get started with outreg2 if you are interested.

<https://www.princeton.edu/~otorres/Outreg2.pdf>

```
ssc install outreg2
```

6.3 Example: Mincer Equations - Replicate Tables

Important Note: Missing values are set to -1 in the CPS PUMS (public-use micro dataset). If you do not account for missing, you will have an incorrect analysis.

First, let's generate some mutually exclusive categories of interest. Recategorize Female:

```
gen female = .
replace female = 0 if peosex == 1
replace female = 1 if peosex == 2
label define female1 0 "Male" 1 "Female"
label values female1 female1
```

Generate Union

```
gen union = .
replace union = 0 if peernlab == 2
replace union = 1 if peernlab == 1
label define union1 0 "Nonunion" 1 "Union"
label values union1 union1
```

Our outcome of interest: Weekly Earnings - *PTERNWA*. *Note:* Documentation says that the data in pternwa imply 2 decimals, so we need to divide by 100. We will also use the variable PRERELG to flag individuals with valid earnings.

```

gen earnings = .
replace earnings = pternwa if prerelg==1
*Divide by 100 for decimals
replace earnings = earnings/100

```

Generate Educational Bins

```

tab peeduca
gen educ = .
*High School Drop Out: from Less than 1st Grade to 12th Grade No Diploma
replace educ = 1 if peeduca >= 31 & peeduca <38
*Graduated High School or GED
replace educ = 2 if peeduca == 39
*Some College
replace educ = 3 if peeduca == 40
*AA Degree: Vocational or Academic
replace educ = 4 if peeduca == 41 | peeduca == 42
*Bachelor's Degree
replace educ = 5 if peeduca == 43
*Advanced Degree: Master's, Professional, or Doctorate
replace educ = 6 if peeduca >= 44 & peeduca <= 46
label define educ1 1 "High School Dropout" 2 "High School Graduate" ///
                  3 "Some College" 4 "Associates (VorA) Degree" ///
                  5 "Bachelor's Degree" 6 "Advanced Degree"
label values educ educ1

```

Caveat with Generating Categorical Variables in Stata Don't do `peeduca >= 44` without `peeduca <= 46` since missing values are very large. So you if do `peeduca >= 44` and missing educational status is “”, then missing observations will get categorized as an advanced degree which will be a measurement error.

Generate Potential Experience

```

gen exp = prtage - 16
gen exp2 = exp*exp

```

You can use local macros for testing models

```

local rhs1 i.educ exp exp2
local rhs2 i.educ exp exp2 i.female
local rhs3 i.educ exp exp2 i.female i.union
local rhs4 i.educ exp exp2 i.female i.union i.hryear4
local rhs5 i.educ exp exp2 i.female i.union i.hryear4 i.peio1icd
*Add interaction between female and union
local rhs6 i.educ exp exp2 i.female##i.union i.hryear4 i.peio1icd

```

Run our regression

```
reg earnings `rhs1', robust
reg earnings `rhs2', robust
reg earnings `rhs3', robust
reg earnings `rhs4', robust
reg earnings `rhs5', robust
reg earnings `rhs6', robust
```

Linear regression	Number of obs	=	248,811
	F(7, 248803)	=	11601.57
	Prob > F	=	0.0000
	R-squared	=	0.2645
	Root MSE	=	.72359

		Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
larnings							
educ							
High School Graduate		.409167	.0061648	66.37	0.000	.3970841	.4212499
Some College		.4095462	.0067562	60.62	0.000	.3963043	.422788
Associates (VorA) Degree		.5171878	.0069999	73.88	0.000	.5034681	.5309075
Bachelor's Degree		.8685378	.0063983	135.75	0.000	.8559973	.8810782
Advanced Degree		1.073213	.0069293	154.88	0.000	1.059631	1.086794
exp		.062107	.0004216	147.31	0.000	.0612807	.0629333
exp2		-.0009771	7.65e-06	-127.80	0.000	-.000992	-.0009621
_cons		5.518003	.0069875	789.70	0.000	5.504308	5.531699

Linear regression	Number of obs	=	248,811
	F(8, 248802)	=	12342.42
	Prob > F	=	0.0000
	R-squared	=	0.3031
	Root MSE	=	.70438

		Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
larnings							
educ							
High School Graduate		.4156047	.0059502	69.85	0.000	.4039425	.4272668
Some College		.4408511	.0065453	67.35	0.000	.4280225	.4536797

Associates (VorA) Degree		.5596453	.0067703	82.66	0.000	.5463758	.572
Bachelor's Degree		.9084709	.0061978	146.58	0.000	.8963234	.920
Advanced Degree		1.121183	.0067401	166.34	0.000	1.107973	1.11
exp		.0613624	.0004132	148.52	0.000	.0605526	.061
exp2		-.0009637	7.50e-06	-128.53	0.000	-.0009784	-.00
female							
Female		-.333084	.0028429	-117.16	0.000	-.338656	-.32
_cons		5.660114	.0069181	818.17	0.000	5.646555	5.6

Linear regression

Number of obs = 248,811
F(9, 248801) = 11129.25
Prob > F = 0.0000
R-squared = 0.3040
Root MSE = .7039

		Coef.	Robust Std. Err.	t	P> t	[95% Conf. Inter	
lnearnings							
educ							
High School Graduate		.4125402	.0059457	69.39	0.000	.4008868	.42
Some College		.4372733	.0065413	66.85	0.000	.4244525	.45
Associates (VorA) Degree		.5552335	.0067633	82.09	0.000	.5419776	.56
Bachelor's Degree		.9050996	.0061986	146.02	0.000	.8929506	.91
Advanced Degree		1.114006	.0067622	164.74	0.000	1.100752	1.1
exp		.0609065	.0004135	147.28	0.000	.060096	.06
exp2		-.0009571	7.50e-06	-127.69	0.000	-.0009718	-.00
female							
Female		-.3322049	.0028417	-116.91	0.000	-.3377745	-.32
union							
Union		.0886408	.0041953	21.13	0.000	.0804181	.09
_cons		5.660998	.0069149	818.66	0.000	5.647445	5.6

Linear regression

Number of obs = 248,811
F(10, 248800) = 10030.61
Prob > F = 0.0000
R-squared = 0.3049

Root MSE = .70345

	lnearnings	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
	educ						
	High School Graduate	.4122138	.0059457	69.33	0.000	.4005605	.4238672
	Some College	.4373993	.0065415	66.86	0.000	.424578	.4502205
	Associates (VorA) Degree	.5545629	.0067628	82.00	0.000	.541308	.5678177
	Bachelor's Degree	.9045909	.0061969	145.97	0.000	.8924451	.9167367
	Advanced Degree	1.113443	.0067577	164.77	0.000	1.100198	1.126688
	exp	.0609184	.0004136	147.29	0.000	.0601077	.0617291
	exp2	-.0009573	7.50e-06	-127.69	0.000	-.000972	-.0009426
	female						
	Female	-.3321621	.0028397	-116.97	0.000	-.3377278	-.3265963
	union						
	Union	.0887879	.004192	21.18	0.000	.0805716	.0970041
	hryear4						
	2024	.0501524	.0028201	17.78	0.000	.044625	.0556798
	_cons	5.636116	.0070549	798.90	0.000	5.622289	5.649943

Linear regression

Number of obs = 248,811
 F(22, 248788) = 5310.25
 Prob > F = 0.0000
 R-squared = 0.3349
 Root MSE = .68812

	lnearnings	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]
	educ					
	High School Graduate	.3840322	.0059144	64.93	0.000	.372
	Some College	.4091063	.0065253	62.70	0.000	.39
	Associates (VorA) Degree	.5233398	.0068026	76.93	0.000	.51
	Bachelor's Degree	.8481448	.0063508	133.55	0.000	.835
	Advanced Degree	1.0828	.0069954	154.79	0.000	1.06

exp		.0550617	.0004126	133.46	0.00
exp2		-.0008699	7.42e-06	-117.31	0.00
female					
Female		-.2669904	.0030367	-87.92	0.00
union					
Union		.1018213	.0042456	23.98	0.00
hryear4					
2024		.0511416	.0027589	18.54	0.00
prmjind1					
NAICS 21: Mining		.4601445	.0215585	21.34	0.00
NAICS 23: Construction		.2414286	.0139472	17.31	0.00
NAICS 31-33: Manufacturing		.2237477	.0137168	16.31	0.00
NAICS 42,44-45: Wholesale & Retail		-.0344484	.0137914	-2.50	0.03
NAICS 22,48-49: Utilities & Transportation		.1267323	.0143908	8.81	0.00
NAICS 51: Information		.2449216	.0178831	13.70	0.00
NAICS 52-53: Financial Activities		.2939349	.0142959	20.56	0.00
NAICS 53,55,56: Professional & Business Serv..		.2228786	.0138884	16.05	0.00
NAICS 61-62: Education & Health		-.0262834	.0136592	-1.92	0.05
NAICS 71-72: Leisure & Hospitality		-.1961534	.0141092	-13.90	0.00
NAICS 81: Other Services		-.1098199	.015223	-7.21	0.00
NAICS 92: Public Administration		.1422022	.0142532	9.98	0.00
_cons		5.638847	.0146119	385.91	0.00

Linear regression

Number of obs	=	248,811
F(23, 248787)	=	5081.12
Prob > F	=	0.0000
R-squared	=	0.3349
Root MSE	=	.68812

		Coef.	Robust Std. Err.	t	P> t
lnearnings					
educ					
High School Graduate		.3841487	.0059155	64.94	0.00
Some College		.4092612	.0065272	62.70	0.00
Associates (VorA) Degree		.5235541	.0068043	76.94	0.00
Bachelor's Degree		.8481453	.0063508	133.55	0.00
Advanced Degree		1.08265	.006998	154.71	0.00

	exp	.0550613	.0004126	133.46	0.000	.054
	exp2	-.0008699	7.42e-06	-117.31	0.000	-.000
	female					
	Female	-.2680058	.0031915	-83.97	0.000	-.274
	union					
	Union	.0964584	.0056709	17.01	0.000	.085
	female#union					
	Female#Union	.0112795	.008181	1.38	0.168	-.00
	hryear4					
	2024	.0511369	.0027589	18.53	0.000	.045
	prmjind1					
	NAICS 21: Mining	.4602906	.0215562	21.35	0.000	.41
	NAICS 23: Construction	.2417802	.0139496	17.33	0.000	.214
	NAICS 31-33: Manufacturing	.2239984	.0137179	16.33	0.000	.197
	NAICS 42,44-45: Wholesale & Retail	-.0342786	.013792	-2.49	0.013	-.061
	NAICS 22,48-49: Utilities & Transportation	.1272795	.0143997	8.84	0.000	.099
	NAICS 51: Information	.2452513	.0178858	13.71	0.000	.210
	NAICS 52-53: Financial Activities	.2942334	.0142996	20.58	0.000	.266
	NAICS 53,55,56: Professional & Business Serv..	.2231206	.0138903	16.06	0.000	.195
	NAICS 61-62: Education & Health	-.02622	.0136591	-1.92	0.055	-.052
	NAICS 71-72: Leisure & Hospitality	-.1959203	.0141108	-13.88	0.000	-.223
	NAICS 81: Other Services	-.1095106	.015225	-7.19	0.000	-.139
	NAICS 92: Public Administration	.1426441	.0142568	10.01	0.000	.114
	_cons	5.639084	.0146121	385.92	0.000	5.61

Use esttab for formatted results: [esttab documentation](#).

```
local rhs4 i.educ exp exp2 i.female i.union i.hryear4
local rhs5 i.educ exp exp2 i.female i.union i.hryear4 i.peio1icd
*Add interaction between female and union
local rhs6 i.educ exp exp2 i.female##i.union i.hryear4 i.peio1icd
```

Use est clear to clear out your prior estimates.

```
est clear
```

We use eststo to save a model to compare it to another.

```

eststo reg1: reg earnings `rhs4'
eststo reg2: reg earnings `rhs5'
eststo reg3: reg earnings `rhs6'

```

Output the results

```

esttab, title (Mincer Equation) r2 se noconstant star(* .10 ** .05 *** .01) ///
b(%10.3f) drop (*peio1icd) wide label.

```

Mincer Equation

	(1)		(2)		(3)	
	OLS1		OLS2		OLS3	
High School Gr~e	0.412***	(0.006)	0.384***	(0.006)	0.384***	(0.006)
Some College	0.437***	(0.007)	0.409***	(0.007)	0.409***	(0.007)
Associates (Vo~e	0.555***	(0.007)	0.523***	(0.007)	0.524***	(0.007)
Bachelor's Deg~e	0.905***	(0.006)	0.848***	(0.006)	0.848***	(0.006)
Advanced Degree	1.113***	(0.007)	1.083***	(0.007)	1.083***	(0.007)
exp	0.061***	(0.000)	0.055***	(0.000)	0.055***	(0.000)
exp2	-0.001***	(0.000)	-0.001***	(0.000)	-0.001***	(0.000)
Female	-0.332***	(0.003)	-0.267***	(0.003)	-0.268***	(0.003)
Union	0.089***	(0.005)	0.102***	(0.005)	0.096***	(0.007)
HRYEAR4=2024	0.050***	(0.003)	0.051***	(0.003)	0.051***	(0.003)
NAICS 11: Agri~e			0.000	(.)	0.000	(.)
NAICS 21: Mining			0.460***	(0.022)	0.460***	(0.022)
NAICS 23: Cons~n			0.241***	(0.014)	0.242***	(0.014)
NAICS 31-33: M~g			0.224***	(0.014)	0.224***	(0.014)
NAICS 42,44-45~a			-0.034**	(0.014)	-0.034**	(0.014)
NAICS 22,48-49~n			0.127***	(0.015)	0.127***	(0.015)
NAICS 51: Info~n			0.245***	(0.017)	0.245***	(0.017)
NAICS 52-53: F~e			0.294***	(0.014)	0.294***	(0.014)
NAICS 53,55,56~B			0.223***	(0.014)	0.223***	(0.014)
NAICS 61-62: E~h			-0.026*	(0.014)	-0.026*	(0.014)
NAICS 71-72: L~i			-0.196***	(0.014)	-0.196***	(0.014)
NAICS 81: Othe~s			-0.110***	(0.015)	-0.110***	(0.015)
NAICS 92: Publ~n			0.142***	(0.015)	0.143***	(0.015)
Female # Union					0.011	(0.009)
Constant	5.636***	(0.007)	5.639***	(0.015)	5.639***	(0.015)
Observations	248811		248811		248811	
R-squared	0.305		0.335		0.335	

Standard errors in parentheses

* p<.10, ** p<.05, *** p<.01

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